Siddharth Chandak | CV

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Education

- **Stanford University** — **Stanford, CA, USA**
  Ph.D., Electrical Engineering — *2021–present (expected June 2026)*
  Advisor: Prof. Nicholas Bambos
- **Stanford University** — **Stanford, CA, USA**
  M.S., Electrical Engineering, GPA: 4.15/4 — *2021–2023*
- **Indian Institute of Technology Bombay** — **Mumbai, India**
  B.Tech., Electrical Engineering, CPI 9.89/10 — *2017–2021*
  Project Advisor: Prof. Vivek S. Borkar
  - President of India Gold Medal
  - Honours in Electrical Engineering & Minor in Computer Science

Research Interests

Multi-Agent Learning and Control: My research focuses on multi-agent systems and games, with an emphasis on developing algorithms for distributed learning of equilibria for unknown systems. I also study problems in game control, where the aim is to steer players toward equilibria that satisfy both the local and global objectives.

Stochastic Approximation: I am interested in developing new frameworks for stochastic approximation schemes which enable the design and analysis of algorithms in optimization and reinforcement learning. In addition, I work on obtaining tighter finite-time guarantees for reinforcement learning algorithms.

Selected Awards

- Centennial Teaching Assistant Award from Stanford University in 2025.
- Stanford Graduate Fellowship from 2021 to 2026.
- President of India Gold Medal at IIT Bombay for the highest GPA among graduating students in 2021.
- Prof. K. C. Mukherjee Award at IIT Bombay for best B.Tech. project among EE students in 2021.
- Among the top 50 candidates in the Indian National Olympiads in Chemistry and Physics and chosen to attend respective Selection Camps for International Olympiads in 2017.

Publications

Submitted Articles

1. A. Kar, **S. Chandak**, R. Singh, S. Sinhahajari, E. Moulines, S. Bhatnagar, and N. Bambos, "Policy Gradient Methods for Non-Markovian Reinforcement Learning", *Submitted to NeurIPS*, 2026.
2. R. Singh, **S. Chandak**, V. S. Borkar, E. Moulines, and N. Bambos, "Regret and Sample Complexity of Online Q-Learning via Concentration of Stochastic Approximation with Time-Inhomogeneous Markov Chains", *Submitted to NeurIPS*, 2026.
3. **S. Chandak**, R. Tamizholi, and N. Bambos, "Last-Iterate Guarantees for Learning in Co-coercive Games", *Submitted to IEEE Conference on Decision and Control (CDC)*, 2026.
4. **S. Chandak**, A. Yadav, A. Ozgur, and N. Bambos, "Heavy-Tailed and Long-Range Dependent Noise in Stochastic Approximation: A Finite-Time Analysis", *Submitted to IEEE Transactions on Automatic Control*.

5. A. Kar, **S. Chandak**, R. Singh, E. Moulines, S. Bhatnagar, and N. Bambos, "High-Probability Bounds for SGD under the Polyak-Łojasiewicz Condition with Markovian Noise", *Submitted to SIAM Journal on Optimization*.

Journal

1. **S. Chandak**, "$O(1/k)$ Finite-Time Bound for Non-Linear Two-Time-Scale Stochastic Approximation", *IEEE Transactions on Automatic Control*, to appear.
2. **S. Chandak**, "Non-Expansive Mappings in Two-Time-Scale Stochastic Approximation: Finite Time Analysis", *SIAM Journal on Control and Optimization*, to appear.
3. **S. Chandak**, I. Bistritz and N. Bambos, "Choose Your Battles: Distributed Learning Over Multiple Tug of War Games", *IEEE Transactions on Automatic Control*, 2026.
4. **S. Chandak**, I. Bistritz and N. Bambos, "Learning to Control Unknown Strongly Monotone Games", *IEEE Transactions on Control of Network Systems*, 2026.
5. **S. Chandak** and V. S. Borkar, "A Concentration Bound for TD(0) with Function Approximation", *Stochastic Systems*, 2026.
6. **S. Chandak**, P. Shah, V. S. Borkar and P. Dodhia, "Reinforcement Learning in Non-Markovian Environments", *Systems and Control Letters*, 2024.
7. **S. Chandak**, V. S. Borkar and H. Dolhare, "A Concentration Bound for LSPE(λ)", *Systems and Control Letters*, 2023.
8. **S. Chandak**, V. S. Borkar and P. Dodhia, "Concentration of Contractive Stochastic Approximation and Reinforcement Learning", *Stochastic Systems*, 2022.
9. S. U. Haque, **S. Chandak**, F. Chiariotti, D. Günduz and P. Popovski, "Learning to Speak on Behalf of a Group: Medium Access Control for Sending a Shared Message," *IEEE Communications Letters*, 2022.
10. V. S. Borkar and **S. Chandak**, "Prospect-theoretic Q-learning", *Systems and Control Letters*, 2021.
11. **S. Chandak**, F. Chiariotti and P. Popovski, "Hidden Markov Model-Based Encoding for Time-Correlated IoT Sources", *IEEE Communications Letters*, 2021.

Conference

1. **S. Chandak**, S. U. Haque and N. Bambos, "Finite-Time Bounds for Two-Time-Scale Stochastic Approximation with Arbitrary Norm Contractions and Markovian Noise", *IEEE Conference on Decision and Control (CDC)*, 2025.
2. **S. Chandak**, I. Thapa, N. Bambos and D. Scheinker, "Optimal Control for Remote Patient Monitoring with Multidimensional Health States", *IEEE International Conference on Communications (ICC)*, 2025.
3. **S. Chandak**, I. Thapa, N. Bambos and D. Scheinker, "Tiered Service Architecture for Remote Patient Monitoring", *IEEE Healthcom*, 2024.
4. **S. Chandak**, I. Bistritz and N. Bambos, "Tug of Peace: Distributed Learning for Quality of Service Guarantees", *IEEE Conference on Decision and Control (CDC)*, 2023.
5. **S. Chandak**, I. Bistritz and N. Bambos, "Equilibrium Bandits: Learning Optimal Equilibria of Unknown Dynamics", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2023.

Invited Talks

- Learning to Control Unknown Strongly Monotone Games
  - Seminar at Automatic Control Lab, EPFL, March 2026
- O(1/k) Finite-Time Bound for Non-Linear Two-Time-Scale Stochastic Approximation
  - INFORMS Annual Meeting, October 2025.
- Learning to Control Unknown Multi-Agent Systems
  - Department of Electrical Engineering Seminar, IIT Bombay, September 2025.
  - STCS Seminar, Tata Institute of Fundamental Research (TIFR) Mumbai, September 2025.

- Non-Expansive Mappings in Two-Time-Scale Stochastic Approximation
  - INFORMS Applied Probability Society Conference, July 2025.

Teaching Experience

Course Instructor at Stanford University

- ENGR 76: Information Science and Engineering — *Winter 2026*
  - Course aimed at introducing undergraduate students at Stanford to the fundamentals behind modern information and communication systems.

Teaching Assistant at Stanford University

- ENGR 76: Information Science and Engineering — *Spring 2024, 2025*
  Instructor: Prof. Ayfer Ozgur
  - Head TA, leading a team of 10+ TAs for a class of around 300 students.
  - Helped restructure the class by redesigning projects, and designing exams and discussion sessions.
  - Awarded the Centennial Teaching Assistant Award for contributions to the course.
- MS&E 130: Information Networks and Services — *Winter 2023, 2025, 2026*
  - Delivered two guest lectures on queueing and scheduling.
- EE 263: Linear Dynamical Systems — *Autumn 2024*
- MS&E 232: Introduction to Game Theory — *Winter 2024*

Teaching Assistant at IIT Bombay

- MA106: Linear Algebra — *Spring 2020*
- MA105: Calculus — *Fall 2019*
- PH108: Basics of Electricity and Magnetism — *Spring 2019*
- PH107: Quantum Physics and Applications — *Fall 2018*

Experience

- **Uber** | *Risk and Trusted Identity* — Jun. - Sep. 2025
  PhD Machine Learning Engineering Intern
  Developed machine learning models for fraud detection.
- **Morgan Stanley** | *Institutional Equities Division* — Jun. - Aug. 2024
  Quantitative Finance Summer Associate
  Developed a statistical factor model for the Central Risk Book desk in Institutional Equities Division.
- **Aalborg University** | *Supervisor - Prof. Petar Popovski* — May - Jul. 2021
  Summer Intern
  Proposed a multi-agent multi-armed bandit based approach for transmission of a shared message over multiple shared channels while avoiding collision.
- **Aalborg University** | *Supervisor - Prof. Petar Popovski* — Apr. - Jul. 2020
  Summer Intern
  Proposed encoding and decoding scheme for transmitting short IoT packets with time correlation across a noisy channel by modeling source dynamics using Hidden Markov Models.
- **Imperial College London** | *Supervisor - Prof. Nick S. Jones* — May - Jul. 2019
  Summer Intern
  Investigated the difference between social networks in UK, ICL and "Hackspace" - a smaller technical community at ICL, by analyzing survey data on friendships, and modeling social networks using Stochastic Block Models.